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Planning and Execution of Recovering Old Completion in a High-Risk SAP Offshore Well with Tubing Parted at Surface Using Snubbing Unit

B Khairliyev, N. Kumar, and N. Upadhyay, Drilling Department, NEB, Abu-Dhabi, UAE

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Traditionally, the application of snubbing units (fig 1) was confined to operations where tubulars were snubbed under pressure. However, their use has recently expanded to encompass a wide range of activities, including completions, re-completions, fishing, milling, TCP gun perforations, velocity strings, sand control (GP/HRWP), frac/pinpoint stimulation, deployment (e.g., Surgifrac), plug and abandon operations, running smart completions, and gravel packs. In the subject well, the snubbing unit was employed for a workover operation, which included fishing operations and the curing of SAP B.

Workover wells are generally categorized into SAP A and SAP B wells. Although the methods for handling these wells may vary, they are typically managed by mobilizing drilling or workover rigs. Recently, snubbing units have been increasingly used to address issues in workover wells. One of the primary advantages of snubbing units is their cost-effectiveness, particularly in terms of mobilization, as they do not require heavy-duty vehicles. This is especially true for marine operations, where snubbing units are both cost-effective and time-saving. For the subject well, the snubbing unit was chosen primarily to reduce the cost of marine mobilization and due to the urgency of the well, as all drilling rigs were scheduled.

Despite the emergency and complex status of the subject well, which included parted tubing, high pressure in the tubing and Annulus A and B, two slickline fish in the tubing string, and a leaking TRSSV control line, a thorough engineering analysis determined that the snubbing unit was capable of performing the required job. As mentioned earlier, the snubbing unit's lower cost of marine mobilization, due to its smaller size, eliminated the need for a large barge.

Introduction

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Figure 1—Snubbing unit after full rig up

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Given the limitations of snubbing units and the unpredictable nature of fishing in workover wells, it was decided that the objective of the snubbing unit intervention would be limited to temporarily suspending the well and attempting to cure SAP B. The old completion string would be retrieved up to the packer depth, and further washover and retrieval of the tail of the old completion string would be left for the next workover job using a drilling or workover rig.

Challenging Well

The subject well is situated offshore on an artificial island cluster and serves as a single oil producer equipped with CIV, 4 GLM, and TRSSSV (fig 2). Drilled in November 2017, the well features a complex design: a 22" hole with 18 5/8" casing down to 2105 feet, a 16" deviated hole with 13 3/8" casing extending to 7100 feet, a 12 1/2" deviated hole with 9 5/8" casing reaching 11790 feet, an 8 1/2" deviated hole with a 7" liner going to 14077 feet, and a 6" horizontal hole drilled to a depth of 18081 feet. According to deviation survey the well is vertical up to 2300 feet MD and has long tangent section from 6000 ft till 12700 ft at 50-55 degree. The well remained active until 2023, when it was shut down due to reservoir issues.

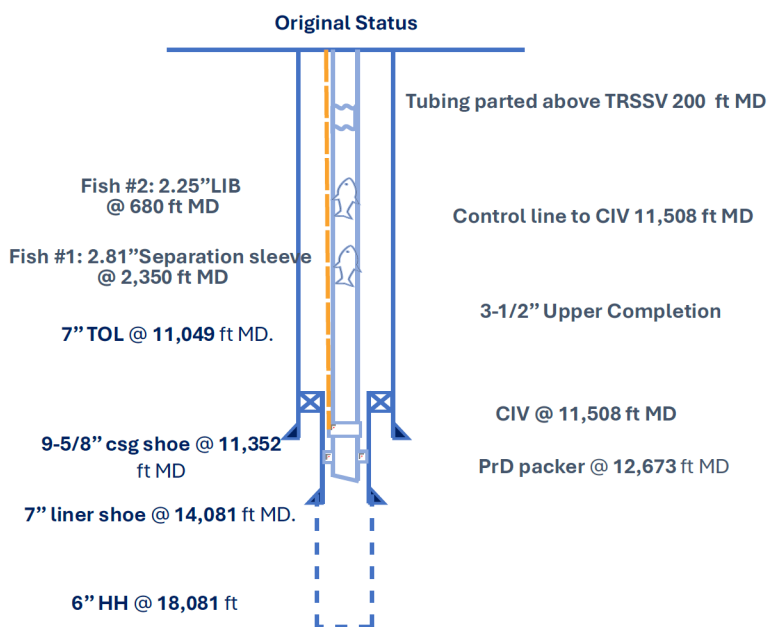


Figure 2—schematic of the well.

During routine maintenance operations, the slickline crew encountered several challenges. They attempted to bleed off the TRSSV control line pressure from 500 psi to zero but were unable to proceed due to gas return. Upon investigation, they discovered a TRSV control line leak and attempted to install a 2.81" separation sleeve, which unfortunately was lost in the hole. Second separation sleeve was installed. Subsequent slickline runs with a 2.00" LIB located the lost separation sleeve at 680 feet. Another slickline run with a 3" GS to retrieve the separation sleeve resulted in the loss of the tool string at 2350 feet. Finally 1 separation sleeve was retrieved but two fish lost in hole.

Fish Summary

The fish summary includes a 2.81" separation sleeve, a 1 7/8" rope socket with a 3' stem bar, KJ, a 5' stem bar, KJ, a mechanical jar, and GS. Another assembly included a 1 7/8" rope socket with KJ, a 2-1/8" 5' stem bar, a 1-7/8" 2' stem bar, a 1-7/8" tubular jar, and a 2.25" LIB.

Attempts to fish using coil tubing revealed that during the accessibility run with a slim BHA and LIB run with size 2.5, the tag was at 1362 ft with an irregular impression. Another LIB run with size 2.75 provided a clearer impression at 1510 ft.

Rigless Wireline Camera Run

Since coil tubing runs with LIB were not conclusive (reached different depth) and considering that TRSSV is not operational in order to understand the situation, it was decided to run wireline camera. Rigless wireline camera run identified suspected parted tubing at 199 ft, with visible TRSSV and CIV control lines in the parted section (fig. 3).

The wellhead pressure readings were 2100 psi in the tubing, 2100 psi in Annulus A, and 1600 psi in Annulus B.

Due to high pressure in annulus A and B, and considering complicated status of the well, the well was considered as emergency well.



Figure 5 DV image of TRSSV control lines in casing

Figure 3—camera run showing 2 control lines CIV and TRSSV.

Planning of the Well

Despite the emergency and complicated status of the subject well, which included parted tubing, high pressure in the tubing and Annulus A and B, two slickline fish in the tubing string, and a leaking Tubing Retrievable Surface Safety Valve (TRSSV) control line, a deep engineering analysis determined that the snubbing unit was capable of performing the required job. As mentioned earlier, the snubbing unit's lower cost of marine mobilization, due to its smaller size, eliminated the need for a large barge.

Simulations showed that a 4" Drill Pipe (DP) could be used for the fishing operation without exceeding buckling limits, and there was still a safe margin for tensile strength. The analysis was performed in Landmark Well plan, using a friction factor of 0.2 for the cased hole since there was no plan to reach the open hole.

One of the limitations of the snubbing unit is that it is not possible to close the Blowout Preventer (BOP) on tubing with control line, which was the case for the subject well. Therefore, the need for killing the well was critical, and this was successfully implemented with the help of the production team.

The fishing program was prepared in collaboration with the Fishing Service provider and included conventional methods of fishing since the 4" DP with XT-39 connection was planned to be used. During simulations, the maximum torque and maximum overpull were limited to the snubbing unit's limitations, which are 460,000 lbs for overpull and 20,000 ft-lbs of torque. The fishing operation was expected to be complicated by parted tubing at top of TRSSV. As a solution two methods were planned – run of overshot and further mechanical back off or use of external cutter.

Through tubing fishing was also considered at the planning stage using a coil tubing unit. However, at a later stage, the use of the coil tubing unit was rejected due to operational and safety restrictions.

Ice Plug

One of the challenges was the lack of barriers to proceed with the nipple down of the Xmas tree. Since the tubing was found parted, pressure testing of the annulus and packer was not possible, and the packer was not considered as a barrier. As a solution, Halliburton Cryogenic Freeze plug was implemented using liquid nitrogen (N₂).

Below is the explanation from Halliburton on Cryogenic Freeze plug technique and design: The freezing method is used when other means of plugging a vessel are impractical, a secondary barrier is required, or bleeding off pressures is impractical. Proper procedures and materials are necessary to perform this service safely. The process involves filling all annuli with a freezable fluid and maintaining this fluid level until it changes to a frozen state within the desired column. When the fluid is frozen and becomes a solid plug, it may be pressure tested up to and exceeding a 10,000-psi differential. N₂ is a common consumable within the oil industry and is typically much more accessible than bulk quantities of dry ice. Loss of N₂ volumes related to mobilization transportation are significantly less than that of dry ice. One of the largest considerations in selecting between dry ice and N₂ is the ability to get N₂ modular tanks within a reasonable proximity to the well or desired freeze point. In the event there are large distances between the N₂ source and the desired freeze point, N₂ may prematurely switch from a liquid to a gas within the transport piping, ultimately resulting in a failed freeze plug (plug will not form). Careful consideration should be taken in keeping the N₂ transport as close to the proposed freeze point as possible. If longer distances between N₂ source and freeze point is required, it is possible to utilize larger diameter freeze tubing to keep the N₂ in a liquid state longer. The spotting point of N₂ sources must be evaluated before the freeze equipment is mobilized to the location and any freeze plug is attempted. The N₂ transport will have to be spotted upwind and within a reasonable distance from the well in a safe atmosphere (not oxygen deficient). The outer casing strings must be capable of transferring the freeze into the tubing. All casing strings and tubing are required to have freezable gel or a thermally conductive material across the freeze area. Cryogenically freezing through multiple strings of casing is technically feasible but will require significantly longer times for the heat in the freezable medium and casing to be extracted by the liquid N₂. Rule of thumb is that it requires 1 hour to freeze across 1 inch. It should be noted that carbon steel should never be taken below -150°F due to crystalline structure change which can impact tubulars under load. This method involved wrapping two freeze coils around the wellhead area and introducing liquid N₂ to cool the steel and internal fluids.

During the freeze, additional seawater was slowly pumped to maintain the freeze fluid across the freeze area and detect the formation of the freeze plug via an increase in pump pressure. Pre-hydrated bentonite fluid was used as the freeze fluid to overcome challenges due to losses after killing the well.

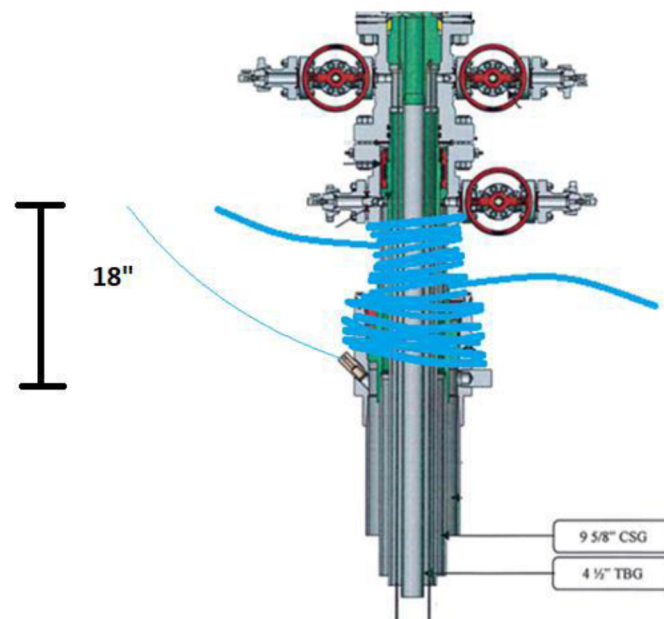


Figure 4—Cryogenic Freeze plug installation schematic. Liquid N₂ pass through wrapped two freeze coils around wellhead area.

Overshot BHA: 8 1/8" Overshot Guide, 8 1/8" Overshot Dressed w 3 1/2" B.G, Overshot Top sub, 6 1/2" Bumper sub, 6 1/2" Fishing Jar, X-OVER, 5" HWDP, X-OVER, 6 1/2" Intensifier, X-OVER, 4" DP to surface.

Technical Constraints and Operations. The internal dimensions of the BHA, along with those of the 4" drill pipe and crossovers, rendered slickline fishing infeasible, since the fish's outer diameter exceeded the BHA's inner diameter. Consequently, twenty fishing BHA runs were carried out, with individual recoveries ranging from 500 to 3,000 feet of fish. Each operation entailed running the overshot BHA, followed by a slickline gauge cutter to verify access, and subsequently punching and cutting the tubing. Ultimately, all sub-assemblies of the old completion string were retrieved, encompassing a total of 20 fishing runs and 20 wireline punch and cut procedures. Using of Snubbing unit in such complicated fishing operation proven to be cost effective compared to rig utilization considering extended fishing operation.

Operational Challenges. A principal challenge was the management of the BHA in conjunction with the snubbing unit. The configuration of the snubbing unit precluded the use of conventional rig tongs with a cathead, necessitating the employment of power tongs. Initial apprehensions regarding the potential collapse of fishing BHA components, particularly wash pipes, when utilizing power tongs were allayed following comprehensive testing, which confirmed the integrity of the BHA make-up and the absence of any collapses.

Moreover, the operation was complicated by the presence of severely corroded tubing. On several occasions, the tubing string parted while being subjected to overpulls of up to 60,000 lbs. Surface inspections confirmed extensive corrosion of the tubing couplings., which was caused probably by CO₂.

Techniques for Access and Retrieval. After engaging the tubing with the overshot BHA and conducting a slickline gauge cutter run to confirm access, the movement of the fish was frequently minimal, often limited to between 200 and 500 feet. One of the reasons was long tangent section after 6000 feet where inclination was 55 degrees. To address this, light jarring and bullheading were systematically applied, and in certain instances, these efforts permitted access to depths of up to 3,000 feet.

Upon the completion of fishing operations, the remaining sections of the old completion string were cut and recovered to a depth of 11,362 feet. Two fish were not retrieved. No further retrieval attempts were made due to time constraints. The well was then temporarily isolated using an acid-soluble cement plug, enabling progression to the subsequent phase: remediation of SAP B.

Setting up pressure control equipment for wireline work also adds extra hours due to the necessity of scaffolding.

Based on the experiences and lessons learned from the recent fishing operation with the snubbing unit, here are some recommendations for future use:

Thorough Pre-Operation Planning: Ensure comprehensive planning and risk assessment before commencing operations. Evaluate all potential risks and challenges and develop contingency plans. This includes assessing the condition of the tubing and other downhole equipment to anticipate issues like corrosion. If possible corrosion log could be run to evaluate the condition of the tubing.

Enhanced Communication and Collaboration: Foster strong communication and collaboration between all teams involved, including operations, engineering, and service providers. Regular meetings and updates can help ensure everyone is aligned and aware of the operation's progress and any emerging challenges.

Use of Advanced Tools and Technology: Invest in advanced tools and technology that can enhance the efficiency and safety of snubbing unit operations. This includes high-quality power tongs, advanced fishing tools, and real-time monitoring systems to track the operation's progress and detect issues early.

Training and Skill Development: Provide ongoing training and skill development for personnel involved in snubbing unit operations. This ensures that the team is well-prepared to handle the unique challenges of snubbing operations and can effectively use the tools and equipment.

Regular Equipment Maintenance and Inspection: Implement a rigorous maintenance and inspection schedule for all equipment used in snubbing operations. This helps identify and address potential issues before they become critical, reducing the risk of equipment failure during operations.

Detailed Documentation and Reporting: Maintain detailed documentation and reporting of all snubbing unit operations. This includes recording the challenges faced, solutions implemented, and outcomes achieved. This information can be invaluable for future operations, helping to identify best practices and areas for improvement.

Continuous Improvement and Feedback Loop: Establish a continuous improvement process that incorporates feedback from all team members. Regularly review and analyze the outcomes of snubbing unit operations to identify lessons learned and opportunities for improvement.

By implementing these recommendations, future snubbing unit operations can be conducted more efficiently and safely, minimizing risks and maximizing the chances of success.

Conclusion

The successful planning and execution of recovering the old completion in a high-risk SAP offshore well using a snubbing unit demonstrate the effectiveness and versatility of this technology in complex well operations. The project faced numerous challenges, including parted tubing, high pressure in the tubing and annuli, and the presence of slickline fish in the tubing string. Despite these obstacles, the engineering analysis and careful planning ensured that the snubbing unit could perform the required tasks efficiently and safely.

The use of the snubbing unit provided significant cost savings, particularly in marine mobilization, and allowed for timely intervention in an emergency well situation. The implementation of the cryogenic freeze plug method and the successful killing of the well further highlight the innovative approaches taken to address the unique challenges of this project.

The fishing operation, although complicated by the design and structure of the snubbing unit and the presence of corroded tubing, was meticulously planned and executed, resulting in the successful retrieval of all sub-assemblies in the old completion string. The collaboration between various teams, including Asset, Production Team, Operations Team, Well Integrity Team, the Fishing service provider, Snubbing unit team, played a crucial role in overcoming the technical difficulties and ensuring the project's success.

In conclusion, this project serves as a testament to the importance of thorough planning, innovative problem-solving, and effective teamwork in the oil and gas industry. The lessons learned and best practices established during this operation will undoubtedly contribute to the success of future projects involving snubbing units and other advanced technologies.